Xunjian Yin

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*I study **self-improving LLM agents** that modify their own code and weights, learn new capabilities without forgetting old ones, and stay reliable on long-horizon tasks. I also build **evaluation** methods that measure what models know and where agents fail.*

Education

Duke University, Ph.D. in Computer Science (Advisors: Shuyan Zhou, Bhuwan Dhingra) *08.2025 – 12.2028 (exp.)*
Peking University, M.S. in Computer Science (Advisor: Xiaojun Wan), GPA: 3.81/4.00 *09.2022 – 06.2025*
Peking University, B.S. in Computer Science, GPA: 3.64/4.00 *09.2018 – 06.2022*

Research Experience

Google DeepMind, Student Researcher (Host: Jinmeng Rao; Team Lead: Claire Cui) *05.2026 – 09.2026*

- **Proactive, lossless context management**: built automatic context management in which a long-horizon agent proactively manages its own working memory with a context language model (CLM) and keeps all history recallable without loss. On Terminal-Bench, it raised history recall to 92.6% and task accuracy to 81.5%, compared with 37.0% and 74.1% for native checkpointing. Also explored KV cache reuse.

Amazon AGI SF Lab, Research Collaborator (Manager: Xiangjun Wang) *10.2025 – 05.2026*

- **Controlled environment changes break strong browser agents**: built BreakingWeb (**NeurIPS 2026**), 519 paired clean and intervened tasks on seven self-hosted websites. The interventions lowered the pass rate of six strong agents by 22.9% on average, compared with 10.0% for humans on a first attempt. In 75% of the failures, the agent declared success although the required change never happened.

Duke University, Ph.D. Research *08.2025 – Present*

- **Weight switching as an agent action**: proposed WeAct, which trains capability-specific LoRA adapters over a frozen backbone and adds adapter selection to the agent's action space. On eight deep-research benchmarks, Qwen3-32B with WeAct improved average accuracy by 6.2% over full fine-tuning with less compute, and by 27.5% over the strongest static LoRA-merging method.
- **Reasoning as flows in representation space** (**ICLR 2026**): modeled LLM reasoning as smooth flows of embedding trajectories, in which logical statements act as local controllers of the flow velocity. Controlled experiments that fix the logic and vary the semantics support the theory.

UC Santa Barbara, NLP Group, Visiting Research Scholar (Advisor: William Yang Wang) *06.2024 – 02.2025*

- **An agent that rewrites its own code**: led Gödel Agent (**ACL 2025**), a self-referential framework in which an LLM agent rewrites its own code at runtime. Guided only by high-level goals, it improved itself continuously. With GPT-3.5, it reached 64.2% on MGSM, compared with 53.4% for Meta Agent Search.
- **Reverse language models**: pre-trained LEDOM (**ACL 2026**), the first purely reverse language model, from scratch at 2B and 7B scale on 435B tokens with previous-token prediction. Reranking QwenMath outputs with LEDOM (Reverse Reward) raised AIME 2024 accuracy from 16.7% with greedy decoding to 23.3%.

Microsoft Research Asia, Research Intern (Advisors: Shuming Ma and Kai Chen) *02.2022 – 08.2022*

- **Curriculum pre-training**: found that curriculum learning speeds up pre-training convergence.

Peking University, Earlier Research (Advisors: Xiaojun Wan, Yunfang Wu, and Sujian Li) *07.2019 – 06.2025*

- **How LLMs handle new and changing knowledge**: built ALCUNA (**EMNLP 2023**), a benchmark of artificial new knowledge, and introduced temporal knowledge editing (**AAAI 2024**), showing that edited models forget historical facts. Defined knowledge boundaries (**ACL 2024**) for prompt-robust evaluation.
- **Text generation and its evaluation**: analyzed Seq2Seq data-to-text models with MQM error annotation (**ACL 2022**) and co-developed Themis (**EMNLP 2024**), a reference-free NLG evaluation model.
- **Chinese spelling check**: proposed RERIC (**LREC-COLING 2024**), a plug-and-play retrieval method that uses phonetic, morphologic, and contextual information.
- **Early projects**: multi-task learning for grammatical error correction (with Yunfang Wu, 2020–2021) and a benchmark of mathematical olympiad problems (with Sujian Li, 2019).

Selected Publications

(equal contribution)*

Preprints:

[arXiv 2026] **WeAct: Dynamic Weight Switching as Agent Actions**
Rui Shi*, **Xunjian Yin***, Jiacheng Sang*, Zichuan Liu, Shuyan Zhou.

[arXiv 2025] **From Atomic to Composite: RL Enables Generalization in Complementary Reasoning**
Sitao Cheng, **Xunjian Yin**, Ruiwen Zhou, ..., Liangming Pan, William Yang Wang, Victor Zhong.

Conference Papers:

[NeurIPS 2026] **Constructing Challenging Browser-Use Tasks by Controlled Environment Interventions**
Xunjian Yin*, Tianchen Guan*, Jinao Wang, ..., Bhuwan Dhingra, Xiangjun Wang, Shuyan Zhou.

[ACL 2025] **Gödel Agent: A Self-Referential Agent Framework for Recursively Self-Improvement**
Xunjian Yin, Xinyi Wang, Liangming Pan, Li Lin, Xiaojun Wan, William Yang Wang.

[ACL 2026] **LEDOM: Reverse Language Model**
Xunjian Yin, Sitao Cheng, Yuxi Xie, ..., Xinyi Wang, Liangming Pan, William Yang Wang, Xiaojun Wan.

[ICLR 2026] **The Geometry of Reasoning: Flowing Logics in Representation Space**
Yufa Zhou*, Yixiao Wang*, **Xunjian Yin***, Shuyan Zhou, Anru R. Zhang.

[ACL 2024] **Benchmarking Knowledge Boundary for LLMs: A Different Perspective on Model Evaluation**
Xunjian Yin*, Xu Zhang*, Jie Ruan, Xiaojun Wan.

[AAAI 2024] **History Matters: Temporal Knowledge Editing in Large Language Model**
Xunjian Yin, Jin Jiang, Liming Yang, Xiaojun Wan.

[EMNLP 2023] **ALCUNA: Large Language Models Meet New Knowledge**
Xunjian Yin*, Baizhou Huang*, Xiaojun Wan.

[ACL 2022] **How Do Seq2Seq Models Perform on End-to-End Data-to-Text Generation?**
Xunjian Yin, Xiaojun Wan.

[EMNLP 2025] **DAMON: A Dialogue-Aware MCTS Framework for Jailbreaking Large Language Models**
Xu Zhang, **Xunjian Yin**, Dinghao Jing, Huixuan Zhang, Xinyu Hu, Xiaojun Wan.

[ICLR 2025] **ChemAgent: Self-updating Memories in LLMs Improves Chemical Reasoning**
Xiangru Tang, Tianyu Hu, Muyang Ye, Yanjun Shao, **Xunjian Yin**, ..., Arman Cohan, Mark Gerstein.

[EMNLP 2024] **Themis: A Reference-free NLG Evaluation Language Model with Flexibility and Interpretability**
Xinyu Hu, Li Lin, Mingqi Gao, **Xunjian Yin**, Xiaojun Wan.

Honors and Awards

Gold Reviewer Award, ICML 2026 *05.2026*
Outstanding Graduate of Beijing; **Outstanding Graduate of Peking University** *06.2025*
Outstanding Master's Thesis Award, WICT, Peking University *06.2025*
Peking University: **Merit Student** and **Guotai Junan Scholarship** (2024), **Wang Xuan Scholarship** (2022), **Award for Research Excellence** (2021–2023), **Award for Academic Excellence** (2019–2020)

Invited Talks

Gödel Agent: A Self-Referential Agent for Recursive Self-Improvement, JetBrains Seminar *09.2025*
Gödel Agent: A Self-Referential Agent for Recursive Self-Improvement, NICE Community *06.2025*
Graduate Speaker, Graduation Ceremony, Peking University *06.2025*

Service

Reviewer: ACL ARR (2023–now), ICLR (2024–2026), NeurIPS (2024–2026), ICML (2025–2026), AAAI (2024–2025), COLM (2025), COLING (2024), JCST (2025)
Teaching Assistant: Natural Language Processing (Duke, Spring 2026, with Bhuwan Dhingra); Web Data Mining (Fall 2023), Introduction to Computing in Python (Fall 2023) and in C++ (Fall 2021), Data Structures and Algorithms (Spring 2022) at Peking University
Volunteer: AAAI 2024, ACL 2024; NLPCC 2023 Shared Task 8 track chair

Skills

Pre-training from scratch (2B–7B), distributed training, efficient inference; Python, C++, PyTorch